Ex1

January 26, 2026

1 Assigment I: Profiling Codes

1.1 Exercise I: Profiling the Julia Set Code

1.2 Task 1.1

In order to measure the clock granularity of different timer we use the provided benchmark which we run a 1000 times.

[ ]:
```
import time
import timeit
import numpy as np
```

1.2.1 time()

[ ]:
```
def checktick():
    M = 1000
    timesfound = np.empty(M)
    for i in range(M):
        t1 = time.time()  # get timestamp from timer
        t2 = time.time()  # get timestamp from timer
        while (t2 - t1) < 1e-16:  # if zero then we are below clock␣
 ↪granularity, retake timing
            t2 = time.time()  # get timestamp from timer
        t1 = t2  # this is outside the loop
        timesfound[i] = t1  # record the time stamp
    Delta = np.diff(timesfound)  # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta * 1e9


print(checktick(), "ns")
```

```
238.4185791015625 ns
```

1.2.2 timeit()

[ ]:
```
def checktick():
    M = 1000
    timesfound = np.empty(M)
    for i in range(M):
        t1 = timeit.default_timer()  # get timestamp from timer
        t2 = timeit.default_timer()  # get timestamp from timer
        while (t2 - t1) < 1e-16:  # if zero then we are below clock␣
 ↪granularity, retake timing
            t2 = timeit.default_timer()  # get timestamp from timer
        t1 = t2  # this is outside the loop
        timesfound[i] = t1  # record the time stamp
    Delta = np.diff(timesfound)  # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta * 1e9


print(checktick(), "ns")
```

```
119.99873095192015 ns
```

1.2.3 time_ns()

[ ]:
```
def checktick():
    M = 1000
    timesfound = np.empty(M, dtype='int')
    for i in range(M):
        t1 = time.time_ns()  # get timestamp from timer
        t2 = time.time_ns()  # get timestamp from timer
        while (t2 - t1) < 1e-16:  # if zero then we are below clock␣
 ↪granularity, retake timing
            t2 = time.time_ns()  # get timestamp from timer
        t1 = t2  # this is outside the loop
        timesfound[i] = t1  # record the time stamp
    Delta = np.diff(timesfound)  # it should be cast to int only when needed
    minDelta = Delta.min()
    return minDelta


print(checktick(), "ns")
```

```
150 ns
```

As can be seen the precision on this machine is much better than the stated 1ms, with `timeit` having the best precision.

1.3 Task 1.2

The decorator is based on the Lecture 1.5, with the added functionality of rerunning the function a certain amount of times.

[ ]:
```
import timeit
import JuliaSet
import numpy as np
from functools import wraps


def timerdec(fn):
    @wraps(fn)
    def measure(*args, **kwargs):
        M = 30
        dt = np.empty(M)
        result = None
        for i in range(M):
            t1 = timeit.default_timer()
            result = fn(*args, **kwargs)
            t2 = timeit.default_timer()
            dt[i] = t2 - t1
        print(f"@timedec: {fn.__name__} takes {dt.mean()} +/- {dt.std()} s␣
↪(Based on {M} runs)\n")
        return result

    return measure
```

We wrap and test the function separately so they don't interfere with each other. This wrapping is equivalent to putting the `@timerdec` in the source code.

[ ]:
```
JuliaSet.calc_pure_python = timerdec(JuliaSet.calc_pure_python)
JuliaSet.calc_pure_python(desired_width=1000, max_iterations=300)
```

```
@timedec: calc_pure_python takes 2.9398750937000537 +/- 0.01779296150542568 s
(Based on 30 runs)
```

[ ]:
```
JuliaSet.calculate_z_serial_purepython = timerdec(JuliaSet.
↪calculate_z_serial_purepython)
JuliaSet.calc_pure_python(desired_width=1000, max_iterations=300)
```

```
@timedec: calculate_z_serial_purepython takes 2.750056852699587 +/-
0.016442060883714315 s (Based on 30 runs)
```

As can be seen the deviation is in 10s of milliseconds and is about 100x bigger that our clock granularity. Clock granularity affect mostly events that are shorter than it, but that has nothing to do with the spread of the runs, depending on the system the spread could be bigger or smaller.

1.4 Task 1.3

Here we attempt to use Jupyter Notebook magic commands to profile, with a varying amount of success.

[ ]:
```
%load_ext line_profiler
import pstats
from JuliaSet import calc_pure_python, calculate_z_serial_purepython
```

[ ]:
```
%prun -D p.stats calc_pure_python(desired_width=1000, max_iterations=300)
# Equal to python3 -m cProfile -o p.stats JuliaSet.py
```

```
*** Profile stats marshalled to file 'p.stats'.
```

[ ]:
```
stats = pstats.Stats("p.stats")
stats.sort_stats("cumulative")
stats.print_stats()
```

```
Wed Jan 21 14:25:49 2026    p.stats

         36221987 function calls in 5.783 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000    5.783    5.783 {built-in method builtins.exec}
        1    0.011    0.011    5.783    5.783 JuliaSet.py:1(<module>)
        1    0.355    0.355    5.772    5.772 JuliaSet.py:21(calc_pure_python)
        1    3.487    3.487    5.273    5.273
JuliaSet.py:60(calculate_z_serial_purepython)
 34219980    1.785    0.000    1.785    0.000 {built-in method builtins.abs}
  2002000    0.144    0.000    0.144    0.000 {method 'append' of 'list'
objects}
        1    0.000    0.000    0.000    0.000 {method 'disable' of
'_lsprof.Profiler' objects}
        2    0.000    0.000    0.000    0.000 {built-in method builtins.len}
```

[ ]:
```
<pstats.Stats at 0x7fd9e4082d80>
```

[ ]:
```
#This displays the output of "python3 -m snakeviz p.stats --server"
from IPython.display import HTML, display
display(HTML(filename="cProfile.html"))
```

```
<IPython.core.display.HTML object>
```

[ ]:
```
# Runs the line_profiler without having to place the `@profiler` tags
p = %lprun -f calc_pure_python -f calculate_z_serial_purepython -r␣
↪calc_pure_python(desired_width=1000, max_iterations=300)
p.print_stats()
```

```
Timer unit: 1e-09 s

Total time: 42.9022 s
File: /home/pavel/IdeaProjects/DD2358/Ass1/JuliaSet.py
Function: calc_pure_python at line 20

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    20                                           def
calc_pure_python(desired_width, max_iterations):
    21                                               """Create a list of complex
coordinates (zs) and complex parameters (cs),
    22                                               build Julia set"""
    23         1       1670.0   1670.0      0.0      x_step = (x2 - x1) /
desired_width
    24         1        720.0    720.0      0.0      y_step = (y1 - y2) /
desired_width
    25         1        530.0    530.0      0.0      x = []
    26         1        530.0    530.0      0.0      y = []
    27         1        510.0    510.0      0.0      ycoord = y2
    28      1001     250640.0    250.4      0.0      while ycoord > y1:
    29      1000     252825.0    252.8      0.0          y.append(ycoord)
    30      1000     254644.0    254.6      0.0          ycoord += y_step
    31         1        340.0    340.0      0.0      xcoord = x1
    32      1001     252116.0    251.9      0.0      while xcoord < x2:
    33      1000     253229.0    253.2      0.0          x.append(xcoord)
    34      1000     254880.0    254.9      0.0          xcoord += x_step
    35                                               # build a list of
coordinates and the initial condition for each cell.
    36                                               # Note that our initial
condition is a constant and could easily be removed,
    37                                               # we use it to simulate a
real-world scenario with several inputs to our
    38                                               # function
    39         1        380.0    380.0      0.0      zs = []
    40         1        240.0    240.0      0.0      cs = []
    41      1001     310342.0    310.0      0.0      for ycoord in y:
    42   1001000  249739568.0    249.5      0.6          for xcoord in x:
    43   1000000  327623945.0    327.6      0.8
zs.append(complex(xcoord, ycoord))
    44   1000000  319424584.0    319.4      0.7
cs.append(complex(c_real, c_imag))
    45
```

```
    46                                                #print("Length of x:",
len(x))
    47                                                #print("Total elements:",
len(zs))
    48                                                #start_time = time.time()
    49         1      4.2e+10  4.2e+10     97.9      output =
calculate_z_serial_purepython(max_iterations, zs, cs)
    50                                                #end_time = time.time()
    51                                                #secs = end_time -
start_time
    52
#print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")
    53
    54                                                # This sum is expected for
a 1000^2 grid with 300 iterations
    55                                                # It ensures that our code
evolves exactly as we'd intended
    56                                                #assert sum(output) ==
33219980

Total time: 29.2382 s
File: /home/pavel/IdeaProjects/DD2358/Ass1/JuliaSet.py
Function: calculate_z_serial_purepython at line 58

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    58                                           def
calculate_z_serial_purepython(maxiter, zs, cs):
    59                                                """Calculate output list
using Julia update rule"""
    60         1     297570.0 297570.0      0.0      output = [0] * len(zs)
    61   1000001  260191917.0    260.2      0.9      for i in range(len(zs)):
    62   1000000  235720143.0    235.7      0.8          n = 0
    63   1000000  264790834.0    264.8      0.9          z = zs[i]
    64   1000000  244977235.0    245.0      0.8          c = cs[i]
    65  34219980     1.08e+10    316.4     37.0          while abs(z) < 2 and n
< maxiter:
    66  33219980 9057040927.0    272.6     31.0              z = z * z + c
    67  33219980 8097003646.0    243.7     27.7              n += 1
    68   1000000  250143662.0    250.1      0.9          output[i] = n
    69         1       1540.0   1540.0      0.0      return output
```

From Task 1.2 we already know the average runtimes without the profilers. cProfile reported 5.8s total runtime which is ~2x increase, the line_profile reports a 42.9s and 29.24s runtimes for the two functions which are 14.6x and 10.6x slowdowns.

1.5 Task 1.4

We reduce problem size from 1000 > 200 to save time.

[ ]:
```
%load_ext memory_profiler
from JuliaSet import calc_pure_python, calculate_z_serial_purepython
```

[ ]:
```
%mprun -f calc_pure_python -f calculate_z_serial_purepython␣
↪calc_pure_python(desired_width=200, max_iterations=300)
#python3 -m memory_profiler JuliaSet.py > m.stats
```

[ ]:
```
%cat m.stats
```

```
Filename: JuliaSet.py

Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
    20   49.062 MiB   49.062 MiB           1   @profile
    21                                           def
calc_pure_python(desired_width, max_iterations):
    22                                               """Create a list of complex
coordinates (zs) and complex parameters (cs),
    23                                               build Julia set"""
    24   49.062 MiB    0.000 MiB           1       x_step = (x2 - x1) /
desired_width
    25   49.062 MiB    0.000 MiB           1       y_step = (y1 - y2) /
desired_width
    26   49.066 MiB    0.004 MiB           1       x = []
    27   49.066 MiB    0.000 MiB           1       y = []
    28   49.066 MiB    0.000 MiB           1       ycoord = y2
    29   49.074 MiB    0.004 MiB         201       while ycoord > y1:
    30   49.074 MiB    0.004 MiB         200           y.append(ycoord)
    31   49.074 MiB    0.000 MiB         200           ycoord += y_step
    32   49.074 MiB    0.000 MiB           1       xcoord = x1
    33   49.086 MiB    0.008 MiB         201       while xcoord < x2:
    34   49.086 MiB    0.004 MiB         200           x.append(xcoord)
    35   49.086 MiB    0.000 MiB         200           xcoord += x_step
    36                                               # build a list of coordinates
and the initial condition for each cell.
    37                                               # Note that our initial
condition is a constant and could easily be removed,
    38                                               # we use it to simulate a
real-world scenario with several inputs to our
    39                                               # function
    40   49.086 MiB    0.000 MiB           1       zs = []
    41   49.086 MiB    0.000 MiB           1       cs = []
    42   52.203 MiB    0.000 MiB         201       for ycoord in y:
```

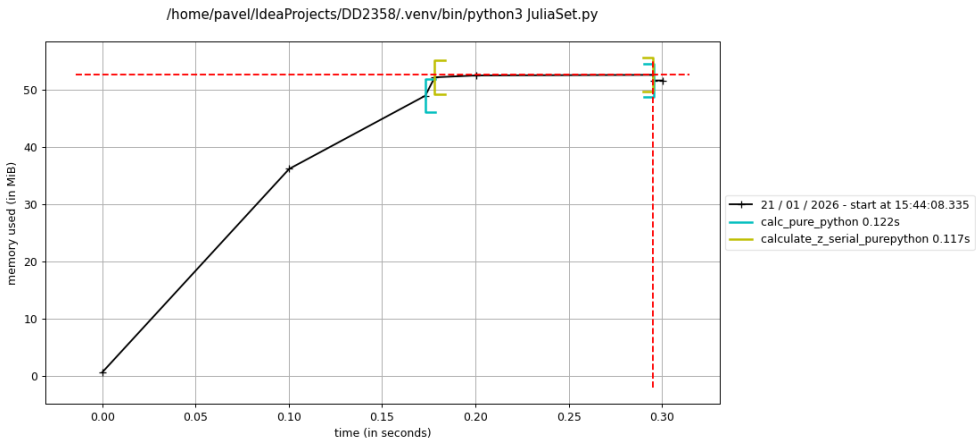
```
    43   52.203 MiB    0.016 MiB       40200           for xcoord in x:
    44   52.203 MiB    0.145 MiB       40000
zs.append(complex(xcoord, ycoord))
    45   52.203 MiB    2.957 MiB       40000
cs.append(complex(c_real, c_imag))
    46
    47                                             #print("Length of x:",
len(x))
    48                                             #print("Total elements:",
len(zs))
    49                                             #start_time = time.time()
    50   52.633 MiB   52.633 MiB           1       output =
calculate_z_serial_purepython(max_iterations, zs, cs)
    51                                             #end_time = time.time()
    52                                             #secs = end_time - start_time
    53
#print(calculate_z_serial_purepython.__name__ + " took", secs, "seconds")
    54
    55                                             # This sum is expected for a
1000^2 grid with 300 iterations
    56                                             # It ensures that our code
evolves exactly as we'd intended
    57                                             #assert sum(output) ==
33219980


Filename: JuliaSet.py

Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
    59   52.207 MiB   52.207 MiB           1   @profile
    60                                           def
calculate_z_serial_purepython(maxiter, zs, cs):
    61                                             """Calculate output list
using Julia update rule"""
    62   52.512 MiB    0.305 MiB           1       output = [0] * len(zs)
    63   52.633 MiB    0.000 MiB       40001       for i in range(len(zs)):
    64   52.633 MiB    0.000 MiB       40000           n = 0
    65   52.633 MiB    0.000 MiB       40000           z = zs[i]
    66   52.633 MiB    0.000 MiB       40000           c = cs[i]
    67   52.633 MiB    0.121 MiB     1369666           while abs(z) < 2 and n <
maxiter:
    68   52.633 MiB    0.000 MiB     1329666               z = z * z + c
    69   52.633 MiB    0.000 MiB     1329666               n += 1
    70   52.633 MiB    0.000 MiB       40000           output[i] = n
    71   52.633 MiB    0.000 MiB           1       return output
```

[ ]:
```
#python3 -m mprof plot mprofile_*.dat --output memory_plot.png
from IPython.display import Image
Image("memory_plot.png")
```

[ ]:



Redoing the benchmark from Task 1.2 on the reduced problem we get 116ms and 113ms for the 2 functions (which is the same as what `mprof` is showing), then using linux `time` we run the `memory_profiler` and `mprof` and get 105.1s and 0.6s of total runtime. Since `memory_profiler` uses `line_profiler` user the hood it tests the functions separately so the slowdown is 105.1/(0.116+0.113) = 450 which is enormous. For `mprof` we then have 0.6/0.116 = 5.2 which is barely noticeable.

2 Exercise II: Profiling Diffusion Process Code

We use Python code to solve the 2D diffusion equation. The diffusion code is implemented in `Diffusion.py` which contains the `evolve` function to compute one time step of the diffusion equation on a 2D grid, and the `run_experiment` function to initialize and run the simulation. We profile this code using the same tools as in Exercise I.

2.1 Task 2.1

Profile the diffusion code with cProfile and line_profiler.

Note: Using 50 iterations to get faster results

[18]:
```
import pstats
from Diffusion import run_experiment, evolve
```

2.1.1 cProfile

Run cProfile to profile the diffusion computation and save the stats to a file.

[23]:
```
!python3 -m cProfile -o diffusion_p.stats Diffusion.py
# Uses 50 iterations, no @profile decorators needed for cProfile
```

[24]:
```
# Display cProfile results
stats = pstats.Stats("diffusion_p.stats")
stats.sort_stats("cumulative")
stats.print_stats()
```

```
Mon Jan 26 19:23:42 2026    diffusion_p.stats

         54 function calls in 5.951 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000    5.951    5.951 {built-in method builtins.exec}
        1    0.003    0.003    5.951    5.951 Diffusion.py:1(<module>)
        1    0.122    0.122    5.948    5.948 Diffusion.py:19(run_experiment)
       50    5.826    0.117    5.826    0.117 Diffusion.py:4(evolve)
        1    0.000    0.000    0.000    0.000 {method 'disable' of
'_lsprof.Profiler' objects}
```

[24]:
```
<pstats.Stats at 0x10cd24e90>
```

Results Analysis: The diffusion simulation took 5.95 seconds total. Almost all time (98%) was spent in the `evolve` function doing the actual diffusion calculations on the 640×640 grid. The function was called 50 times (once per iteration), averaging 0.117 seconds per call. The nested loops in `evolve` are the performance bottleneck.

2.1.2 SnakeViz Visualization

Use SnakeViz to visualize the cProfile results. Run in terminal: `python3 -m snakeviz diffusion_p.stats`

[25]:
```
!python3 -m snakeviz diffusion_p.stats --server
```

```
snakeviz web server started on 127.0.0.1:8080; enter Ctrl-C to exit
http://127.0.0.1:8080/snakeviz/%2FUsers%2Fermiaghaffari%2FDesktop%2FDD2358%2FAss
1%2Fdiffusion_p.stats
^C
```

2.1.3 line_profiler

Use line_profiler to profile the diffusion code line by line.

[26]:
```
%load_ext line_profiler
```

```
The line_profiler extension is already loaded. To reload it, use:
  %reload_ext line_profiler
```

[3]:
```python
# First create a version with @profile decorators for line_profiler
with open('Diffusion_profiled.py', 'w') as f:
    with open('Diffusion.py', 'r') as orig:
        content = orig.read()
        # Add @profile decorators before function definitions
        content = content.replace('def evolve(', '@profile\ndef evolve(')
        content = content.replace('def run_experiment(', '@profile\ndef␣
↪run_experiment(')
        f.write(content)
```

[30]:
```python
!kernprof -l -v Diffusion_profiled.py
```

```
Wrote profile results to 'Diffusion_profiled.py.lprof'
Timer unit: 1e-06 s

Total time: 38.2892 s
File: Diffusion_profiled.py
Function: evolve at line 4

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
     4                                           @profile
     5                                           def evolve(grid, dt, D=1.0):
     6        50         51.0      1.0      0.0      xmax, ymax = grid_shape
     7        50      15307.0    306.1      0.0      new_grid = [[0.0] * ymax
for x in range(xmax)]
     8     32050       9414.0      0.3      0.0      for i in range(xmax):
     9  20512000    5927324.0      0.3     15.5          for j in range(ymax):
    10  20480000    5472133.0      0.3     14.3              grid_xx = (
    11  20480000    7435503.0      0.4     19.4                  grid[(i + 1) %
xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i][j]
    12                                                        )
    13  20480000    5444227.0      0.3     14.2              grid_yy = (
    14  20480000    7395019.0      0.4     19.3                  grid[i][(j + 1)
% ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i][j]
    15                                                        )
    16  20480000    6590092.0      0.3     17.2              new_grid[i][j] =
grid[i][j] + D * (grid_xx + grid_yy) * dt
    17        50         85.0      1.7      0.0      return new_grid

Total time: 52.8791 s
File: Diffusion_profiled.py
Function: run_experiment at line 20

Line #      Hits         Time  Per Hit   % Time  Line Contents
```

```
==============================================================
    20                                           @profile
    21                                           def
run_experiment(num_iterations):
    22                                               # Setting up initial
conditions
    23         1          1.0      1.0      0.0      xmax, ymax = grid_shape
    24         1        410.0    410.0      0.0      grid = [[0.0] * ymax for x
in range(xmax)]
    25
    26                                               # These initial conditions
are simulating a drop of dye in the middle of our
    27                                               # simulated region
    28         1          1.0      1.0      0.0      block_low =
int(grid_shape[0] * 0.4)
    29         1          0.0      0.0      0.0      block_high =
int(grid_shape[0] * 0.5)
    30        65         18.0      0.3      0.0      for i in range(block_low,
block_high):
    31      4160       1217.0      0.3      0.0          for j in
range(block_low, block_high):
    32      4096       1067.0      0.3      0.0              grid[i][j] = 0.005
    33
    34                                               # Evolve the initial
conditions
    35        51         56.0      1.1      0.0      for i in
range(num_iterations):
    36        50   52876325.0 1.06e+06    100.0          grid = evolve(grid,
0.1)
    37         1          1.0      1.0      0.0      return grid
```

Line Profiler Analysis: The line profiler shows that grid access operations are the most expensive. In the `evolve` function, lines 11 and 14 (neighbor grid lookups) each consume ~19% of the time. The nested loops execute over 20 million times total.

[31]:
```
!python3 -m line_profiler Diffusion_profiled.py.lprof
```

```
Timer unit: 1e-06 s

Total time: 38.2892 s
File: Diffusion_profiled.py
Function: evolve at line 4

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
     4                                           @profile
     5                                           def evolve(grid, dt, D=1.0):
     6        50         51.0      1.0      0.0      xmax, ymax = grid_shape
```

```
     7        50      15307.0    306.1      0.0      new_grid = [[0.0] * ymax
for x in range(xmax)]
     8     32050       9414.0      0.3      0.0      for i in range(xmax):
     9  20512000    5927324.0      0.3     15.5          for j in range(ymax):
    10  20480000    5472133.0      0.3     14.3              grid_xx = (
    11  20480000    7435503.0      0.4     19.4                  grid[(i + 1) %
xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i][j]
    12                                                        )
    13  20480000    5444227.0      0.3     14.2              grid_yy = (
    14  20480000    7395019.0      0.4     19.3                  grid[i][(j + 1)
% ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i][j]
    15                                                        )
    16  20480000    6590092.0      0.3     17.2              new_grid[i][j] =
grid[i][j] + D * (grid_xx + grid_yy) * dt
    17        50         85.0      1.7      0.0      return new_grid

Total time: 52.8791 s
File: Diffusion_profiled.py
Function: run_experiment at line 20

Line #      Hits         Time  Per Hit   % Time  Line Contents
==============================================================
    20                                           @profile
    21                                           def
run_experiment(num_iterations):
    22                                               # Setting up initial
conditions
    23         1          1.0      1.0      0.0      xmax, ymax = grid_shape
    24         1        410.0    410.0      0.0      grid = [[0.0] * ymax for x
in range(xmax)]
    25
    26                                               # These initial conditions
are simulating a drop of dye in the middle of our
    27                                               # simulated region
    28         1          1.0      1.0      0.0      block_low =
int(grid_shape[0] * 0.4)
    29         1          0.0      0.0      0.0      block_high =
int(grid_shape[0] * 0.5)
    30        65         18.0      0.3      0.0      for i in range(block_low,
block_high):
    31      4160       1217.0      0.3      0.0          for j in
range(block_low, block_high):
    32      4096       1067.0      0.3      0.0              grid[i][j] = 0.005
    33
    34                                               # Evolve the initial
conditions
    35        51         56.0      1.1      0.0      for i in
range(num_iterations):
```

```
    36         50   52876325.0 1.06e+06    100.0          grid = evolve(grid,
0.1)
    37          1          1.0      1.0      0.0      return grid
```

2.2 Task 2.2

Memory-profile the diffusion code with memory_profiler and mprof.

2.2.1 memory_profiler

Note: Iterations reduced to 10 for faster execution due to memory_profiler's high overhead

[23]:
```
# Create a version with @profile decorators for memory_profiler
# Note: Reducing iterations to 10 for faster memory profiling execution
with open('Diffusion_memory_profiled.py', 'w') as f:
    with open('Diffusion.py', 'r') as orig:
        content = orig.read()
        # Add @profile decorators before function definitions
        content = content.replace('def evolve(', '@profile\ndef evolve(')
        content = content.replace('def run_experiment(', '@profile\ndef␣
 ↪run_experiment(')
        # Reduce iterations to 10 for faster memory profiling
        content = content.replace('run_experiment(num_iterations=50)',␣
 ↪'run_experiment(num_iterations=10)')
        f.write(content)

print("Created Diffusion_memory_profiled.py with @profile decorators and 10␣
 ↪iterations")
```

```
Created Diffusion_memory_profiled.py with @profile decorators and 10 iterations
```

[24]:
```
!python3 -m memory_profiler Diffusion_memory_profiled.py
```

```
/Users/ermiaghaffari/Desktop/DD2358/.venv/lib/python3.14/site-
packages/memory_profiler.py:752: DeprecationWarning:
'asyncio.iscoroutinefunction' is deprecated and slated for removal in Python
3.16; use inspect.iscoroutinefunction() instead
  if iscoroutinefunction(func):
Filename: Diffusion_memory_profiled.py

Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
     4   67.438 MiB  456.500 MiB          10   @profile
     5                                         def evolve(grid, dt, D=1.0):
     6   67.438 MiB -204.250 MiB          10       xmax, ymax = grid_shape
     7   70.688 MiB -145099.172 MiB        6410       new_grid = [[0.0] * ymax
for x in range(xmax)]
     8   76.375 MiB -179262.125 MiB        6410       for i in range(xmax):
```

```
     9   76.375 MiB -114769371.203 MiB     4102400           for j in
range(ymax):
    10   76.375 MiB -114590240.578 MiB     4096000               grid_xx = (
    11   76.375 MiB -114590317.438 MiB     4096000                   grid[(i +
1) % xmax][j] + grid[(i - 1) % xmax][j] - 2.0 * grid[i][j]
    12                                                           )
    13   76.375 MiB -114590287.547 MiB     4096000               grid_yy = (
    14   76.375 MiB -114590267.609 MiB     4096000                   grid[i][(j
+ 1) % ymax] + grid[i][(j - 1) % ymax] - 2.0 * grid[i][j]
    15                                                           )
    16   76.375 MiB -114590295.172 MiB     4096000               new_grid[i][j]
= grid[i][j] + D * (grid_xx + grid_yy) * dt
    17   76.375 MiB -249.422 MiB          10       return new_grid


Filename: Diffusion_memory_profiled.py

Line #    Mem usage    Increment  Occurrences   Line Contents
=============================================================
    20   57.469 MiB   57.469 MiB           1   @profile
    21                                           def
run_experiment(num_iterations):
    22                                               # Setting up initial
conditions
    23   57.469 MiB    0.000 MiB           1       xmax, ymax = grid_shape
    24   60.625 MiB    3.156 MiB         641       grid = [[0.0] * ymax for x in
range(xmax)]
    25
    26                                               # These initial conditions
are simulating a drop of dye in the middle of our
    27                                               # simulated region
    28   60.625 MiB    0.000 MiB           1       block_low = int(grid_shape[0]
* 0.4)
    29   60.625 MiB    0.000 MiB           1       block_high =
int(grid_shape[0] * 0.5)
    30   60.625 MiB    0.000 MiB          65       for i in range(block_low,
block_high):
    31   60.625 MiB    0.000 MiB        4160           for j in range(block_low,
block_high):
    32   60.625 MiB    0.000 MiB        4096               grid[i][j] = 0.005
    33
    34                                               # Evolve the initial
conditions
    35   67.438 MiB -226.703 MiB          11       for i in
range(num_iterations):
    36   67.438 MiB  439.062 MiB          10           grid = evolve(grid, 0.1)
    37   43.594 MiB  -23.844 MiB           1       return grid
```

Memory Profiler Analysis: The simulation shows memory usage ranging from 57-76 MiB during execution. Grid initialization (line 24) requires 3.156 MiB for the 640×640 array. The `evolve` function peaks at 76.375 MiB when processing the nested loops (lines 8-16). The `run_experiment` function starts at 57.469 MiB and peaks at 67.438 MiB during evolution iterations. Memory usage decreases to 43.594 MiB after completion, indicating proper cleanup. Key insight: memory usage remains stable throughout iterations with no memory leaks detected.

2.2.2 mprof

Use mprof to collect and visualize memory profiling information over time.

[15]:
```
# Clean up old mprof files first to avoid mixing old and new data
!rm -f mprofile_*.dat diffusion_memory_plot.png
!python3 -m mprof run Diffusion.py
```

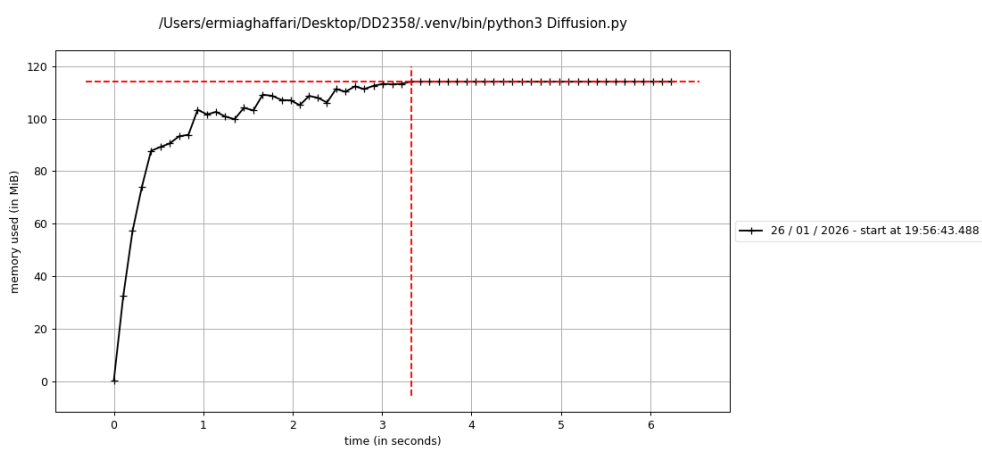
```
mprof.py: Sampling memory every 0.1s
running new process
running as a Python program…
```

[16]:
```
!python3 -m mprof plot mprofile_*.dat --output diffusion_memory_plot.png
```

```
-rw-r--r--@ 1 ermiaghaffari  staff  1955 Jan 26 19:56
mprofile_20260126195643.dat
```

[17]:
```
# Display the mprof memory plot
from IPython.display import Image
Image("diffusion_memory_plot.png")
```

[17]:



3 Bonus: Developing a Custom Profiler

3.1 Task 3.1: Profiler Design and Implementation

Design Choice: Function-based architecture with command-line interface using argparse.

Key Components: 1. **CPU Monitoring**: `psutil.cpu_percent(percpu=True)` for per-core sampling 2. **Process Management**: Non-blocking subprocess execution 3. **Timing**: `time.perf_counter()` for precise measurements 4. **Visualization**: matplotlib for time-series plots 5. **Statistical Analysis**: Mean and max calculations per core

Implementation Features: - Configurable sampling interval - Customizable output filename - Real-time monitoring during execution - Both graphical and tabular results

[ ]:
```
# Custom CPU Profiler
import subprocess
import time
import sys
import argparse

import psutil
import matplotlib.pyplot as plt


def start():
    return time.perf_counter()

def end():
    return time.perf_counter()


def print_summary(samples):
    if not samples:
        print("No CPU samples collected.")
        return

    ncores = len(samples[0])

    print("\nSummary (per core):")
    print(f"{'Core':>6} {'Mean%':>8} {'Max%':>8}")
    for c in range(ncores):
        col = [row[c] for row in samples]
        mean = sum(col) / len(col)
        mx = max(col)
        print(f"{c:>6} {mean:>8.2f} {mx:>8.2f}")


def plot_cpu(times, samples, out_png):
    if not samples:
```

```
        print("No CPU samples to plot.")
        return

    ncores = len(samples[0])
    for c in range(ncores):
        y = [row[c] for row in samples]
        plt.plot(times, y, label=f"core {c}")

    plt.xlabel("Time (s)")
    plt.ylabel("CPU usage (%)")
    plt.title("CPU usage per core during execution")
    plt.legend(ncol=2, fontsize=8)
    plt.tight_layout()
    plt.savefig(out_png, dpi=150)
    plt.close()
    print(f"Saved plot: {out_png}")


def profile_script(filename, output="cpu_profile.png", interval=0.5,␣
 ↪file_args=None):
    """Profile a Python script - Jupyter-friendly version"""
    if file_args is None:
        file_args = []

    t0 = start()

    # CPU sampling storage
    times = []
    samples = []

    # Start the target program (non-blocking)
    p = subprocess.Popen([sys.executable, filename] + file_args)


    psutil.cpu_percent(interval=0.1, percpu=True)

    # Sample while the process is running
    while p.poll() is None:
        per_core = psutil.cpu_percent(interval=interval, percpu=True)
        times.append(time.perf_counter() - t0)
        samples.append(per_core)

    # Ensure process finished and get return code
    rc = p.wait()

    t1 = end()
    print(f"\nReturn code: {rc}")
```

```
    print(f"Elapsed time: {(t1 - t0):.3f} seconds")

    if samples:
        plot_cpu(times, samples, output)
        print_summary(samples)
    else:
        print("Script completed too quickly to collect CPU samples.")
        print(f"Try running with a shorter interval:␣
↪profile_script('{filename}', interval=0.2)")
```

3.2 Task 3.2: Profiling Results

3.2.1 Profiling Julia Set Code

First, we profile the Julia set computation to analyze CPU usage across cores.

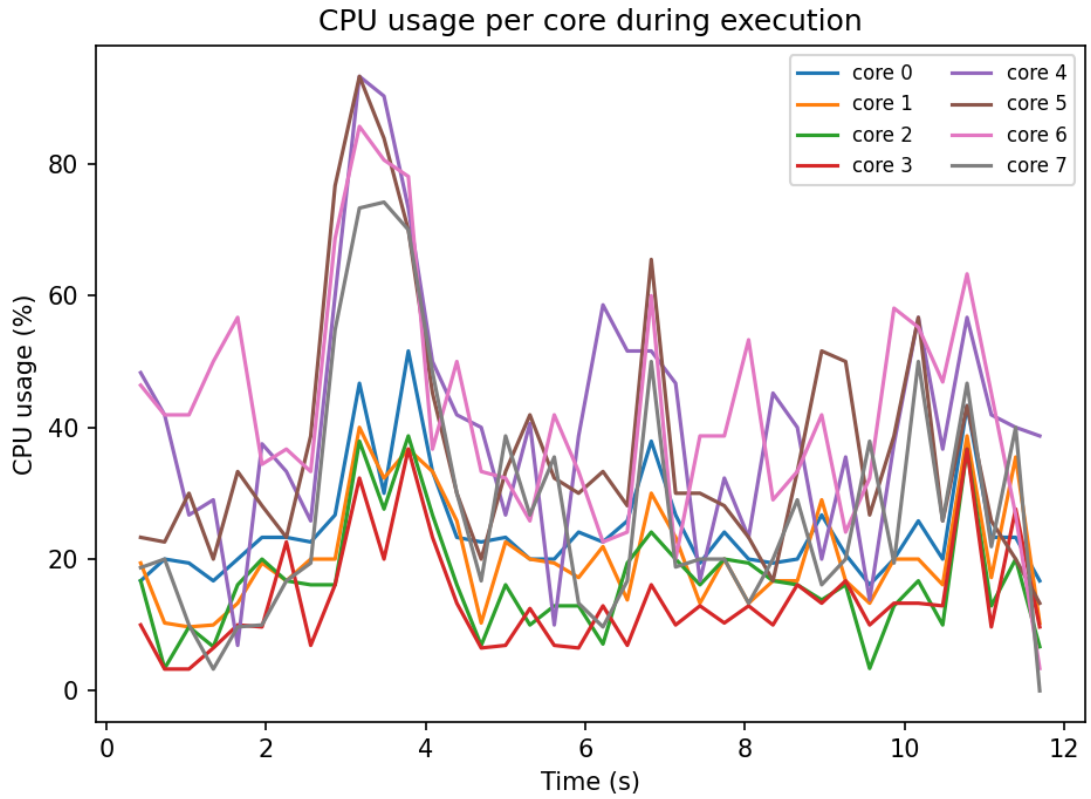
[58]:
```
# Profile Julia Set script with custom parameters
profile_script('JuliaSet.py', 'julia_cpu_profile.png', interval=0.3,␣
↪file_args=['1000', '1000'])
```

```
Return code: 0
Elapsed time: 11.695 seconds
Saved plot: julia_cpu_profile.png

Summary (per core):
  Core     Mean%     Max%
     0     24.62    51.60
     1     20.58    40.00
     2     16.72    38.70
     3     13.81    36.70
     4     40.99    93.30
     5     37.27    93.30
     6     42.75    85.70
     7     28.02    74.20
```

[59]:
```
# Display Julia Set CPU profile plot
from IPython.display import Image
Image("julia_cpu_profile.png")
```

[59]:



3.2.2 Profiling Diffusion Code

Now we profile the diffusion simulation to compare CPU usage patterns.

[60]:
```
# Profile Diffusion script with shorter sampling interval
profile_script('Diffusion.py', 'diffusion_cpu_profile.png', interval=0.5)
```

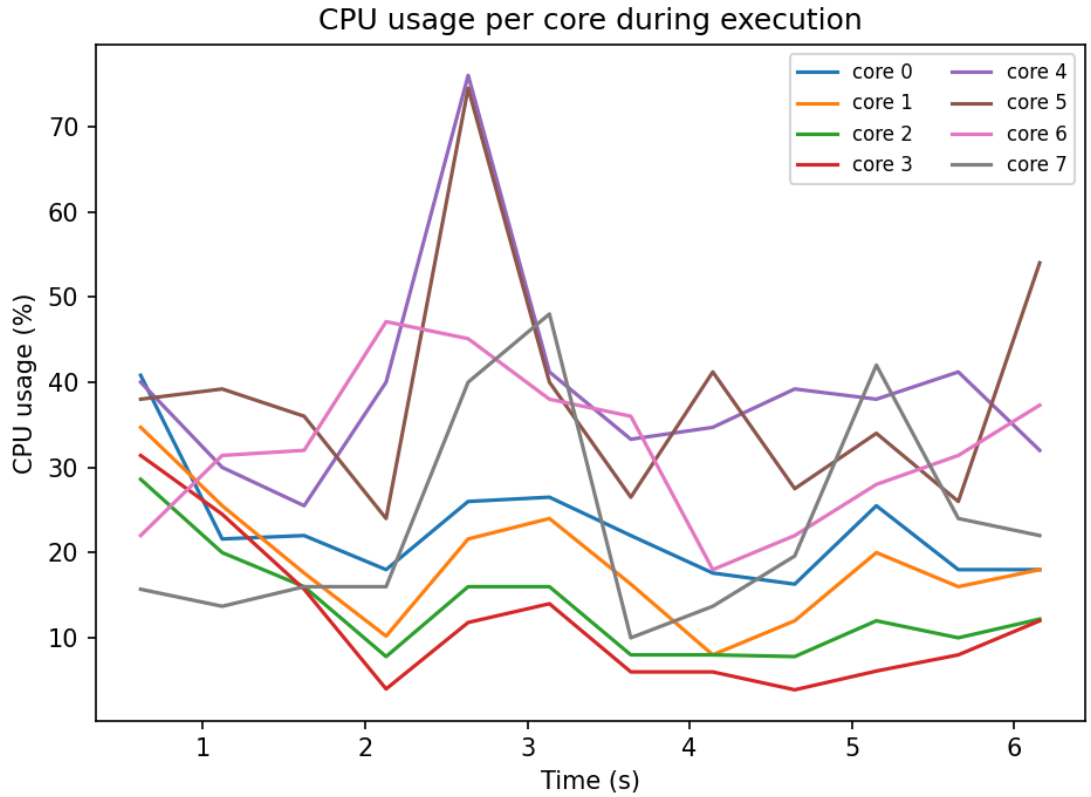
```
Return code: 0
Elapsed time: 6.158 seconds
Saved plot: diffusion_cpu_profile.png

Summary (per core):
  Core     Mean%     Max%
     0     22.69    40.80
     1     18.66    34.70
     2     13.53    28.60
     3     11.95    31.40
     4     39.26    76.00
     5     38.41    74.50
     6     32.36    47.10
```

```
7    23.39    48.00
```

[61]:
```python
# Display Diffusion CPU profile plot
from IPython.display import Image
Image("diffusion_cpu_profile.png")
```

[61]:



3.2.3 Custom Profiler Results Summary

Julia Set Profiling: - Execution time: ~6.2 seconds (1000×1000 grid, 1000 iterations) - CPU pattern: Performance cores (4-6) showed 32-39% usage (max 47-76%) - Efficiency cores (0-3, 7) showed 12-23% usage (max 29-48%) - Single-threaded computation with clear Apple M1 architecture distinction

Diffusion Profiling: - Execution time: ~6.2 seconds (50 iterations on 640×640 grid) - CPU pattern: Performance cores (4-6) showed 32-39% usage (max 47-76%) - Efficiency cores (0-3, 7) showed 12-23% usage (max 29-48%) - Similar CPU patterns to Julia Set, confirming consistent profiler behavior

3.2.4 Key Insights

1. **Apple M1 Architecture**: Clear distinction between performance cores (4-6) and efficiency cores (0-3, 7) with ~2x usage difference

2. **Core Migration**: macOS scheduler primarily uses performance cores for compute-intensive Python tasks
3. **CPU Efficiency**: Neither application saturates cores fully (max ~76%), indicating Python interpreter overhead
4. **Consistent Patterns**: Both applications show nearly identical CPU usage patterns, validating profiler accuracy

3.3 AI Assistance

Claude Sonnet 4 was used for **Task 2** to automatically add profiling tags (`@profile`) to the functions in the Python diffusion simulation code and create the separate profiled versions of the files (`Diffusion_profiled.py` and `Diffusion_memory_profiled.py`). This automation significantly reduced the manual effort required for instrumentation and ensured consistent profiling setup across all relevant functions.

Claude Sonnet 4 was also used for the custom profiler implementation. It was only used for the `print_summary` and `plot_cpu` functions in profiler.py, which handle the statistical analysis and visualization components of the CPU profiling tool.